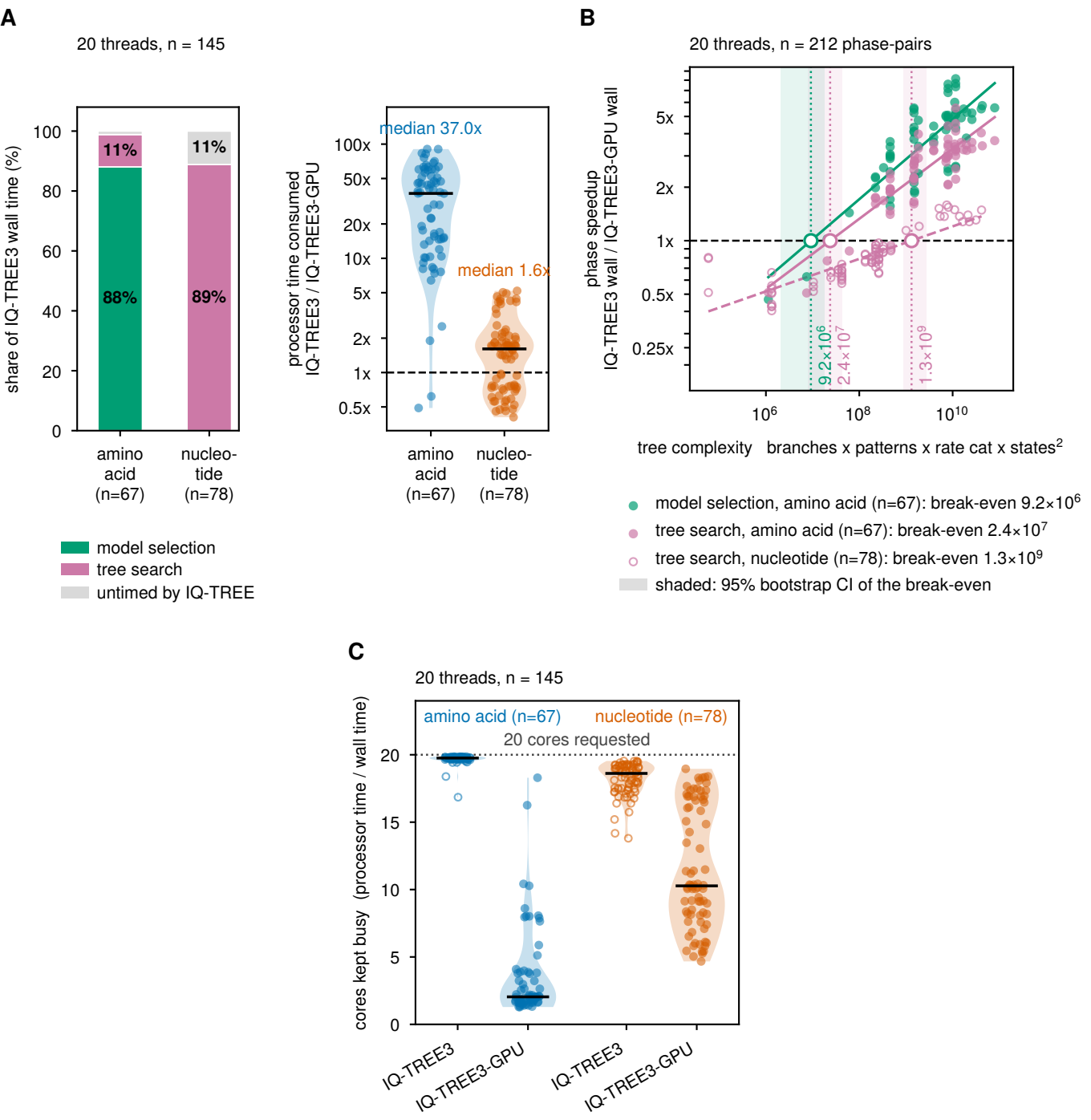

A
20 threads, n = 145
share of IQ-TREE3 wall time (%)
11%
11%
88%
89%
amino acid (n=67)
nucleo-tide (n=78)
model selection
tree search
untimed by IQ-TREE
processor time consumed IQ-TREE3 / IQ-TREE3-GPU
median 37.0x
median 1.6x
B
20 threads, n = 212 phase-pairs
phase speedup IQ-TREE3 wall / IQ-TREE3-GPU wall
9.2×10^6
2.4×10^7
1.3×10^9
tree complexity branches x patterns x rate cat x states^2
model selection, amino acid (n=67): break-even 9.2×10^6
tree search, amino acid (n=67): break-even 2.4×10^7
tree search, nucleotide (n=78): break-even 1.3×10^9
shaded: 95% bootstrap CI of the break-even
C
20 threads, n = 145
amino acid (n=67)
nucleotide (n=78)
20 cores requested
cores kept busy (processor time / wall time)
IQ-TREE3
IQ-TREE3-GPU
IQ-TREE3
IQ-TREE3-GPU